

# PROBLEM SET 3—SOLUTIONS

## ECON301—ECONOMETRICS I

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These are my suggested solutions. It is your responsibility to cross-check this document with your own answers.

1. By the CLT we know that  $Z_n \rightarrow_d N(0, 1)$ . We also have that, since  $E(W_i) = 0$

$$Z_n = \frac{\frac{1}{n} \sum_{i=1}^n W_i}{\sigma_W / \sqrt{n}} = \frac{\frac{1}{\sqrt{n}} \sum_{i=1}^n W_i}{\sigma_W}.$$

Thus,

$$\frac{\frac{1}{\sqrt{n}} \sum_{i=1}^n W_i}{\sigma_W} \rightarrow_d N(0, 1).$$

Using Slutsky's Theorem (Thm. 4.6, ii) we have then that

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n W_i \rightarrow_d \sigma_W N(0, 1) = N(0, \sigma_W^2).$$

An alternative derivation:

$$\Pr\left(\frac{1}{\sqrt{n}} \sum_{i=1}^n W_i \leq w\right) = \Pr\left(\frac{1}{n} \sum_{i=1}^n W_i \leq \frac{w}{\sqrt{n}}\right) = \Pr\left(Z_n \leq \frac{w/\sqrt{n}}{\sigma_W/\sqrt{n}}\right) = \Pr(Z_n \leq w/\sigma_W) \approx \Phi(w/\sigma_W).$$

Therefore,

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n W_i \overset{A}{\rightsquigarrow} N(0, \sigma_W^2).$$

2. (a) Recall that if  $Y_i$  is a Bernoulli random variable with parameter  $p$ , we have that  $E(Y_i) = p$  and  $\text{Var}(Y_i) = p(1-p)$ . Therefore, since  $\{Y_i\}$  is a sequence of i.i.d. random variables,  $E(\hat{p}) = p$  and  $\text{Var}(\hat{p}) = p(1-p)/n$ . Therefore, the sequence

$$Z_n = \frac{\hat{p} - p}{\sqrt{\frac{p(1-p)}{n}}}$$

satisfies the conditions of the CLT and we can conclude that  $Z_n \rightarrow_d N(0, 1)$ .

- (b) We propose the following estimator for  $\text{Var}(Y_1)$ :

$$\widehat{\text{Var}}(Y_1) = \hat{p}(1 - \hat{p}).$$

By the WLLN, we know that  $\hat{p} \rightarrow_P p$ . Define  $g(y) = y(1-y)$ . Then  $g$  is a continuous function in all of its domain and in light of this, we can apply theorem 4.3 to conclude that

$$\widehat{\text{Var}}(Y_1) = \hat{p}(1 - \hat{p}) = g(\hat{p}) \rightarrow_P g(p) = p(1-p) = \text{Var}(Y_1).$$

Therefore, we  $\widehat{\text{Var}}(Y_1)$  is a consistent estimator.

- (c) Let  $T_n$  be defined as

$$T_n = \frac{\hat{p} - p}{\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}},$$

and let  $Z_n$  be defined as in item (a). Firstly, note that  $\sqrt{\hat{p}(1-\hat{p})} \rightarrow_P \sqrt{p(1-p)}$ , i.e.  $\sqrt{\widehat{\text{Var}}(Y_1)}$  is a consistent estimator of  $\sqrt{p(1-p)}$  where, once again, this was an application of theorem 4.3. Define  $W_n$  as

$$W_n = \frac{\sqrt{\hat{p}(1-\hat{p})}}{\sqrt{p(1-p)}}.$$

Then we have that

$$T_n = \frac{\hat{p} - p}{\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}} = \frac{\hat{p} - p}{\frac{\sqrt{p(1-p)}}{\sqrt{n}} \frac{\sqrt{\hat{p}(1-\hat{p})}}{\sqrt{p(1-p)}}} = \frac{Z_n}{W_n},$$

since  $W_n \rightarrow_P 1 \neq 0$ , we have from theorem 4.6 (iii)

$$T_n \rightarrow_d \frac{N(0, 1)}{1} = N(0, 1).$$

Thus,  $T_n$  converges in distribution to a standard normal distribution when  $n$  tends to infinity.

(d) Firstly, we have that

$$\hat{p} = \frac{39}{127} \approx 0.3071, \quad \sqrt{\frac{\hat{p}(1-\hat{p})}{127}} \approx 0.0409.$$

The large sample approximation for the confidence interval is

$$\Pr(|T_n| \leq z_{\alpha/2}) = \Pr\left(\left|\frac{\hat{p} - p}{\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}}\right| \leq z_{\alpha/2}\right) \approx 1 - \alpha.$$

After some simple algebraic manipulations, we obtain:

$$\Pr\left(\hat{p} - z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \leq p \leq \hat{p} + z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}\right) \approx 1 - \alpha.$$

For a 99% confidence interval,  $\alpha = 0.01$  so that  $z_{0.005} = 2.57583$ . Thus, the 99% confidence interval from the large sample approximation is given by

$$[0.3071 \pm 2.57583(0.0409)] = [0.2017, 0.4125].$$

The above is a random interval that will have an *approximate* probability of 99% of containing the true value of  $p$ .

3. We will first show that  $X_n \rightarrow_{L_2} c$ , which by theorem 4.1 implies that  $X_n \rightarrow_P c$ . We have that

$$\begin{aligned} E[(T_n - c)^2] &= E[(T_n - E(T_n) + E(T_n) - c)^2] = E[(T_n - E(T_n))^2 + 2(T_n - E(T_n))(E(T_n) - c) + (E(T_n) - c)^2] \\ &= E[(T_n - E(T_n))^2] + 2(E(T_n) - E(T_n))(E(T_n) - c) + (E(T_n) - c)^2 \\ &= \text{Var}(T_n) + (E(T_n) - c)^2 \end{aligned}$$

Taking limits as  $n$  tends to infinity and using hypothesis we have that  $E[(T_n - c)^2] \rightarrow 0$ . Therefore, we conclude that  $T_n \rightarrow_{L_2} c$  and by theorem 4.1, the convergence is also in probability.

4. Let  $g(X) = (X - \mu)^2 \geq 0$ . Then apply Chebychev's inequality (as in the lecture) with  $r = k^2\sigma^2$  to get

$$\Pr(|X - \mu| \geq k\sigma) = \Pr((X - \mu)^2 \geq k^2\sigma^2) \leq \frac{E[(X - \mu)^2]}{k^2\sigma^2} = \frac{\sigma^2}{k^2\sigma^2} = \frac{1}{k^2}$$

5. We know that  $E[\bar{X}] = \mu$  and  $\text{Var}(\bar{X}) = \sigma_{\bar{X}}^2 = \sigma^2/n = 9/100$  so that  $\sigma_{\bar{X}} = \sqrt{9/100} = 0.3$ .

(a) Using Chebychev's second inequality, we get

$$\Pr(-0.3k < \bar{X} - \mu < 0.3k) \geq 1 - 1/k^2.$$

We need  $1 - 1/k^2 \geq 0.9$  which implies that  $k \geq \sqrt{10} = 3.1623$ . Thus, we can take  $k = 3.1623$ . This implies that

$$\Pr(-0.9487 < \bar{X} - \mu < 0.9487) \geq 0.9.$$

(b) Using the CLT,  $\bar{X} \sim AN(\mu, \sigma_{\bar{X}}^2)$ . and  $(\bar{X} - \mu)/0.3 \sim AN(0, 1)$ . Thus

$$\Pr\left(-1.645 < \frac{\bar{X} - \mu}{0.3} < 1.645\right) = \Pr(-0.4935 < \bar{X} - \mu < 0.4935) \approx 0.9.$$

Thus, the limits obtained by the CLT are tighter than the ones obtained by Chebychev's Second Inequality. In fact, the ones obtained by Chebychev's are almost twice as big as the ones normal approximation. Moreover, with a sample size of 100,  $\bar{X}$  is probably very close to normally distributed, even if the underlying  $X$  distribution is not close to normal. On the other hand, one reason that Chebychev's Inequality is widely applicable to any positive random variable with finite mean and variance is that the assumptions required to use it are quite mild, but the trade-off for this wide applicability is that the bounds obtained with it are quite conservative, i.e., not very tight.